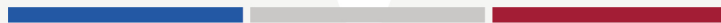




MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING



MODULE 1 UNIT 1
Enabling organizational value

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Learning outcome:

LO1: Determine how agentic AI characteristics enable organizational value creation compared to reactive LLM systems.

1. Introduction

This lesson establishes the foundation for the program by answering a central leadership question: **What is agentic AI, and why does it matter?**

As you progress through the module, you will move from broad familiarity to a clearer, more practical understanding of how agentic systems work, where they create value, and what leaders need to evaluate, adopt, and govern them well.

2. Defining agentic AI

In this video, Paul McDonagh-Smith places agentic AI on the broader AI continuum, from classical machine learning to deep learning to generative AI. He argues that, for all its fluency, generative AI is still reactive: you prompt it, and it responds.

Agentic AI marks a more significant shift. Instead of simply answering a question, it can pursue an objective by planning steps, using tools, maintaining context, and adjusting course when needed. McDonagh-Smith offers an apt analogy: a large language model is like a research assistant, while an agentic system is more like a capable project manager.

He also advises caution. These systems are not truly autonomous yet. They remain brittle, require guardrails, and still need human oversight. Their value lies in amplifying human judgment across multi-step workflows, not replacing it.

Main message: As a leader, it is important to understand where your organization's real opportunities lie on the AI continuum and build accordingly.



Video 1: Defining agentic AI. (Access this lesson on the Online Campus to engage with this video and download its transcript.)

3. From large language models to agents

McDonagh-Smith expands on what gives agentic AI its operational power. He shows that the model itself is only part of the picture. What matters is the surrounding architecture that enables it to gather information, break work into stages, carry context across a process, and check its own output.

He also draws **a sharper line between assistance and execution**. Once AI can query systems, trigger actions, or move work forward on its own, the stakes change. Errors no longer remain contained within a conversation; they can affect real decisions and operations.

Main message: For leaders, the key question is not whether AI can do more, but where greater autonomy is useful, and where it must be constrained.



Video 2: LLM to agents. (Access this lesson on the Online Campus to engage with this video and download its transcript.)

4. The implications of agentic AI

Now, McDonagh-Smith shifts from definition to implication, asking what agentic AI means for how organizations compete and create value. He argues that its significance lies not only in efficiency, but in **changing the economics of knowledge work** and expanding what teams can accomplish.

McDonagh-Smith emphasizes the following three realities leaders should keep in mind:

1. Iteration speed matters more than a perfect plan.
2. The biggest barriers are organizational, not technical.
3. Existing KPIs may not capture the new forms of value these systems create.

The real opportunity is not simply to automate existing processes, but to rethink workflows from scratch.

Main message: The leadership task is not just to adopt agentic AI, but to redesign workflows, capabilities, and metrics to unlock value.



Video 3: Agentic AI implications. (Access this lesson on the Online Campus to engage with this video and download its transcript.)

5. Agentic AI agents

Kanchana Patlolla, a Product Solutions Manager at Google, joins McDonagh-Smith to discuss how quickly the field is changing. Capabilities once associated with agents are increasingly being incorporated into the models themselves, shifting the real challenge toward orchestration, infrastructure, and deployment.

Main message: AI use is evolving from standalone applications to being embedded directly within models.



Video 4: Agentic AI agents. (Access this lesson on the Online Campus to engage with this video and download its transcript.)

Patlolla highlights the key trends leaders should monitor as agentic AI continues to evolve, specifically the rise of specialized, collaborative systems. She also emphasizes the increasing autonomy of agents as they create and work together.

Leadership lesson: Build organizations that can move quickly, learn continuously, and adapt as agentic AI evolves.



Video 5: Swarm of agents. (Access this lesson on the Online Campus to engage with this video and download its transcript.)

Note:

At this point in the lesson, you can test your understanding of the content with a practice quiz. Access this lesson on the Online Campus to begin.

6. Conclusion

Module 1 establishes the foundation for the rest of the program by clarifying **what agentic AI is, how it works, and why it matters for leaders**. The central opportunity is not simply to use more advanced AI tools, but to understand where agentic systems can create meaningful business value and where they require clear guardrails, thoughtful workflow design, and stronger leadership judgment.

Key takeaways:

- Agentic AI marks a shift from systems that respond to prompts to systems that can work toward an objective.
- Its value comes from how it supports multi-step workflows, not just single-task outputs.

The leadership challenge is to decide where autonomy adds value, where it adds risk, and how to design accordingly.